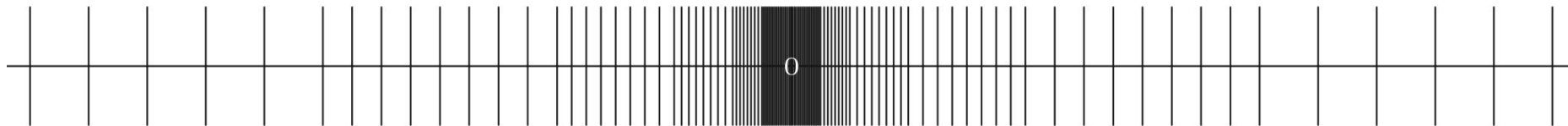


Tools to Detect and Diagnose Floating-Point Errors in Heterogeneous Computing Hardware and Software

A Half-Day Tutorial at Supercomputing 2025

Tut 122: Sunday, 16 November 2025, 8:30am - 12:00pm CST : **Location 121**



Tools to Detect and Diagnose Floating-Point Errors in Heterogeneous Computing Hardware and Software

A Half-Day Tutorial at Supercomputing 2025

Presenters

Ignacio Laguna



Mark Baranowski and
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Cindy Rubio-González

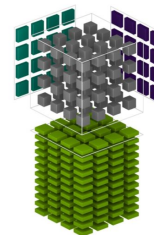
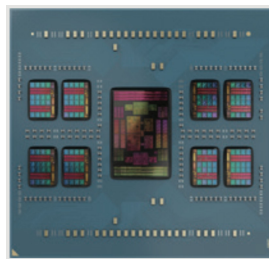


Agenda, and a single QR code to access all our material

- 08:30 : Introduction
- 08:35 : FPChecker
- 09:35 : NixNan (Part-1)
- 09:55 : Fill Survey
- 10:00 : Coffee Break
- 10:30 : Brief Recap
- 10:35 : NixNan (Part-2)
- 11:05 : FloatGuard
- 11:50 : Closing Remarks
- 11:55 : Fill Survey



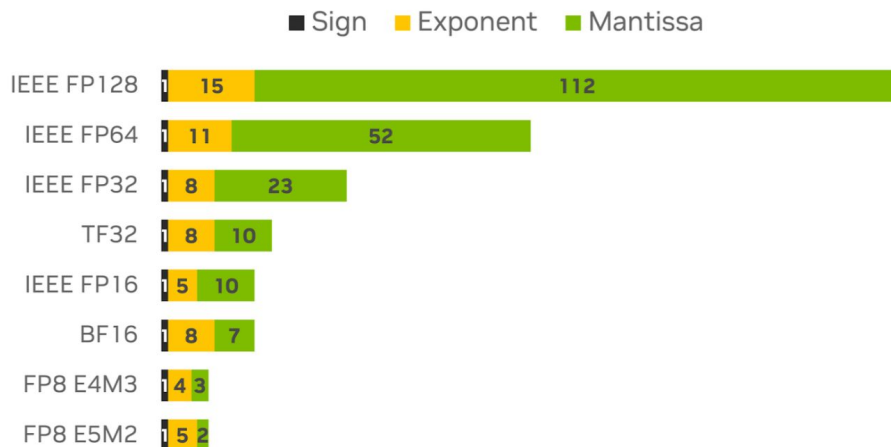
Floating-Point Computations are fundamental to CPUs and GPUs



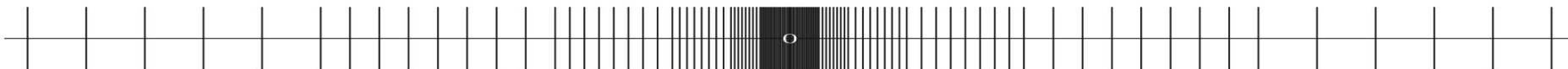
$$D = \begin{matrix} \text{FP16 or FP32} & \text{FP16} & \text{FP16} & \text{FP16 or FP32} \end{matrix} + \begin{matrix} \text{FP16} & \text{FP16} & \text{FP16} & \text{FP16 or FP32} \end{matrix}$$

Floating-Point Numbers : A compromise, with many good properties, that can span a large range with a small number of bits
 Half the representable numbers lie between -1 and +1

Format	Prec.	Exp.	Sm. norm.	Lrgst. Norm.	ULP @ emax	eq. 2's bits	Redn. factor
FP16 16 bits	10	5	6.10 E-5	6.55 E4	5	41	2.5
BF16 16 bits	7	8	1.18 E-38	3.39 E38	120	262	16.3
TF32 19 bits	10	8	1.18 E-38	3.40 E38	117	265	13.9
FP32 32 bits	23	8	1.18 E-38	3.40 E38	104	254	7.9
FP64 64 bits	52	11	2.23 E-308	1.80 E308	971	2099 bits	32.7



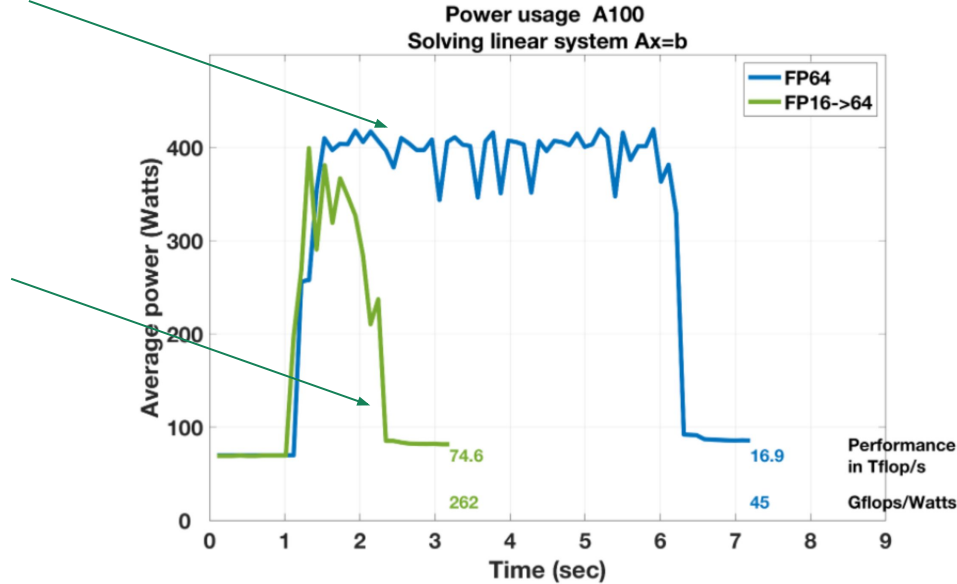
FP allows us to get away moving **64 bits** instead of moving **2099 bits** (in 2's comp.)



Lower precision Floating-Point saves **energy** (and **time**)

High-Precision
LU Solving

Low-Precision
LU Solving



From "Hardware Trends Impacting Floating-Point Computations In Scientific Applications," Dongarra et al.

While increasing **precision** almost always improves **accuracy**, there are rare cases where this does not happen.

- The loop in FP32 iterates 10 times (as per real semantics)
- In FP64, it iterates 11 times

```
File Edit Options Buffers Tools C Help
#include <stdio.h>

int main() {
    int cnt=0;
    for (float i = 0.0f; i < 1.0f; i += 0.1) {
        cnt++;
        printf("%.1f ", i);
    }
    printf("\n");
    printf("cnt=%d\n", cnt);
    return 0;
}
-UU-:--- F1 floatCnt.c Top L5 (C/*l Abbrev) -----
#include <stdio.h>

int main() {
    int cnt=0;
    for (double i = 0.0f; i < 1.0f; i += 0.1) {
        cnt++;
        printf("%.1f ", i);
    }
    printf("\n");
    printf("cnt=%d\n", cnt);
    return 0;
}
-UU-:--- F1 doubCnt.c Top L1 (C/*l Abbrev) -----
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$ gcc -o fc floatCnt.c
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$ gcc -o dc doubCnt.c
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$ ./fc
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
cnt=10
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$ ./dc
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
cnt=11
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$
```

Non-intuitive behavior caused by Rounding!

```
File Edit Options Buffers Tools C Help
#include <stdio.h>

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cnt=11
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$
```

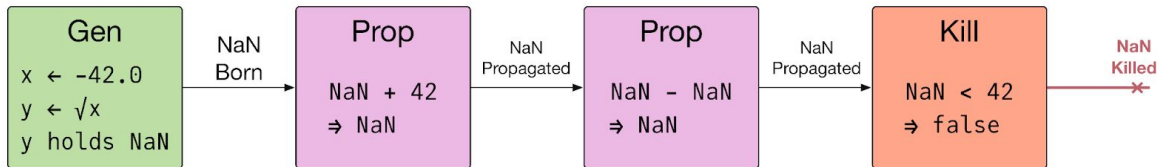
```
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$ ./fc
Value: 0.000000000000000000000000
(sign: 0, exp: 0, mantissa: 0x000000)
Value: 0.1000000014901161193848
(sign: 0, exp: 123, mantissa: 0x4CCCCD)
Value: 0.2000000029802322387695
(sign: 0, exp: 124, mantissa: 0x4CCCCD)
Value: 0.3000000119209289550781
(sign: 0, exp: 125, mantissa: 0x19999A)
Value: 0.4000000059604644775391
(sign: 0, exp: 125, mantissa: 0x4CCCCD)
Value: 0.500000000000000000000000
(sign: 0, exp: 126, mantissa: 0x000000)
Value: 0.6000000238418579101562
(sign: 0, exp: 126, mantissa: 0x19999A)
Value: 0.7000000476837158203125
(sign: 0, exp: 126, mantissa: 0x333334)
Value: 0.8000000715255737304688
(sign: 0, exp: 126, mantissa: 0x4CCCCD)
Value: 0.9000000953674316406250
(sign: 0, exp: 126, mantissa: 0x666668)
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$ ./dc
Value: 0.000000000000000000000000
(sign: 0, exp: 0, mantissa: 0x000000000000)
Value: 0.1000000000000000005511
(sign: 0, exp: 1019, mantissa: 0x999999999999A)
Value: 0.20000000000000000011022
(sign: 0, exp: 1020, mantissa: 0x999999999999A)
Value: 0.30000000000000000444089
(sign: 0, exp: 1021, mantissa: 0x3333333333334)
Value: 0.40000000000000000222045
(sign: 0, exp: 1021, mantissa: 0x999999999999A)
Value: 0.500000000000000000000000
(sign: 0, exp: 1022, mantissa: 0x000000000000)
Value: 0.599999999999999977955
(sign: 0, exp: 1022, mantissa: 0x3333333333334)
Value: 0.699999999999999955911
(sign: 0, exp: 1022, mantissa: 0x6666666666666)
Value: 0.799999999999999933866
(sign: 0, exp: 1022, mantissa: 0x9999999999999)
Value: 0.899999999999999911822
(sign: 0, exp: 1022, mantissa: 0xCCCCCCCCCCCCC)
Value: 0.9999999999999999889777
(sign: 0, exp: 1022, mantissa: 0xFFFFFFFFFFF)
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$
(base) ganesh@ganesh-XPS-17-9710:~/parfloat-cs6961/c-prec-expts$
```


Focus of this tutorial

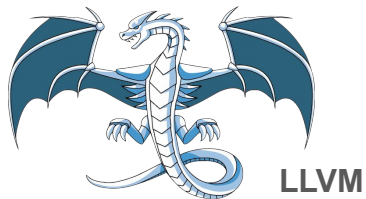
- Excessive Rounding Error (not addressed in this tutorial)
- Exhausting the range (addressed)
 - **Detection of Exceptions (main focus)**

FP Exceptions (Exceptional Values) are Undesirable (*often* are bugs)

- FP32: (1.9×10^{19})² $\rightarrow$ INF ... (in E4M3, $16^2 = \text{INF}$) $\Leftarrow$ a popular ML low-prec format
 - Easily generated during dot-product
- Of the FP Exceptions, we focus on INF and NaN
 - NaNs can skew control-flow
- This macro is buggy
 - `#define MAX(x,y) ( (x >= y) ? x : y )`
hint : consider `x == NaN` or `y == NaN`
- Exceptions often vanish harmlessly (e.g. in ML codes)
 - But it is difficult to know which control-flows they have affected meanwhile

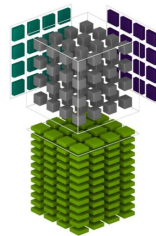


Tools Presented



GPU-FPX + NixNaN

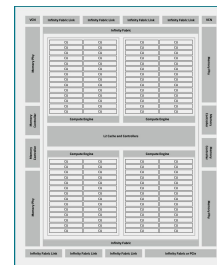
NVIDIA GPUs



$$D = \begin{bmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{bmatrix} + \begin{bmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{bmatrix}$$

FP16 or FP32 FP16 FP16 FP16 or FP32

FloatGuard



AMD
GPUs



 Lawrence Livermore
National Laboratory



 KAHLERT SCHOOL OF COMPUTING
THE UNIVERSITY OF UTAH



 UC DAVIS
UNIVERSITY OF CALIFORNIA



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(Switch to the FPChecker Slides)